

Modulus and Argument of a Complex Number

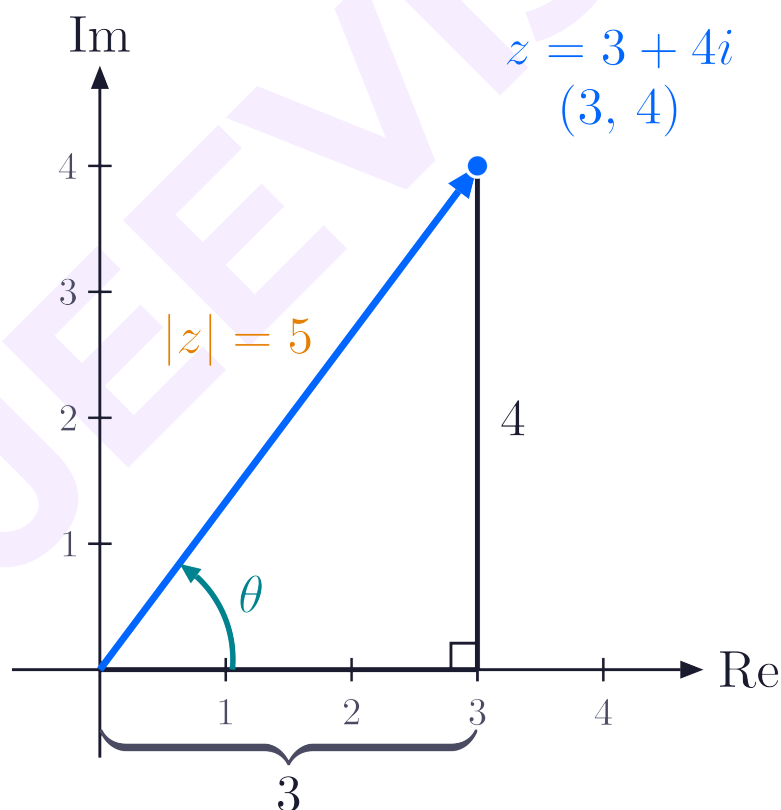


Mathematics | JEE Main & Advanced

A complex number is a point, and also an arrow from the origin. Its modulus is how long the arrow is. Its argument is which way the arrow points.

A complex number is a point and an arrow

The complex number $z = x + iy$ is plotted as the point (x, y) on the Argand plane: the real part x along the real axis (Re) and the imaginary part y along the imaginary axis (Im). Join the origin to that point and you get an arrow. Two numbers describe the arrow completely: its **length** and its **direction**.



For $z = 3 + 4i$ the arrow and the two sides 3 and 4 form a right triangle.

Modulus: how long the arrow is

By Pythagoras, the length of the arrow is

$$|z| = \sqrt{x^2 + y^2}, \quad |3 + 4i| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

The modulus is a length, so it is never negative, and it is 0 only for $z = 0$.

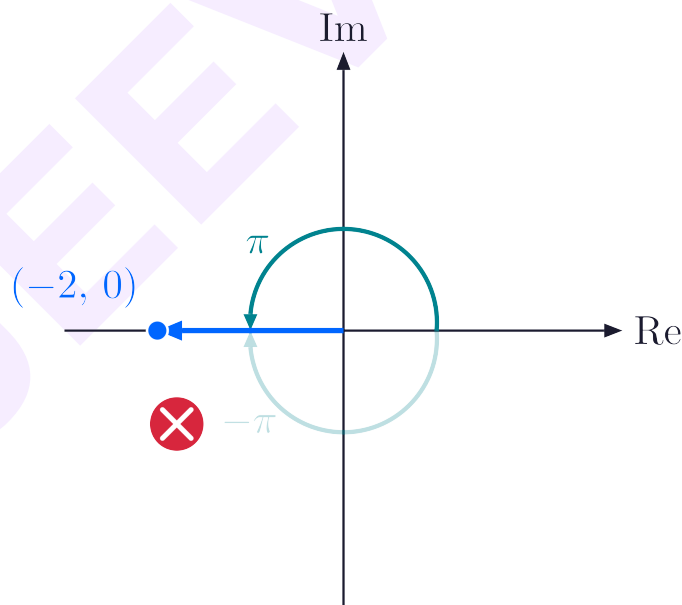
Argument: which way the arrow points

The argument $\arg z$ is the angle from the positive real axis to the arrow. Anticlockwise turns count as positive and clockwise turns as negative. For $3 + 4i$, $\tan \theta = \frac{4}{3}$, so $\theta = \tan^{-1}\left(\frac{4}{3}\right) \approx 0.9273$ radians, which is about 53.13° . For $z = 0$ there is no arrow, so $\arg 0$ is not defined.

The principal argument

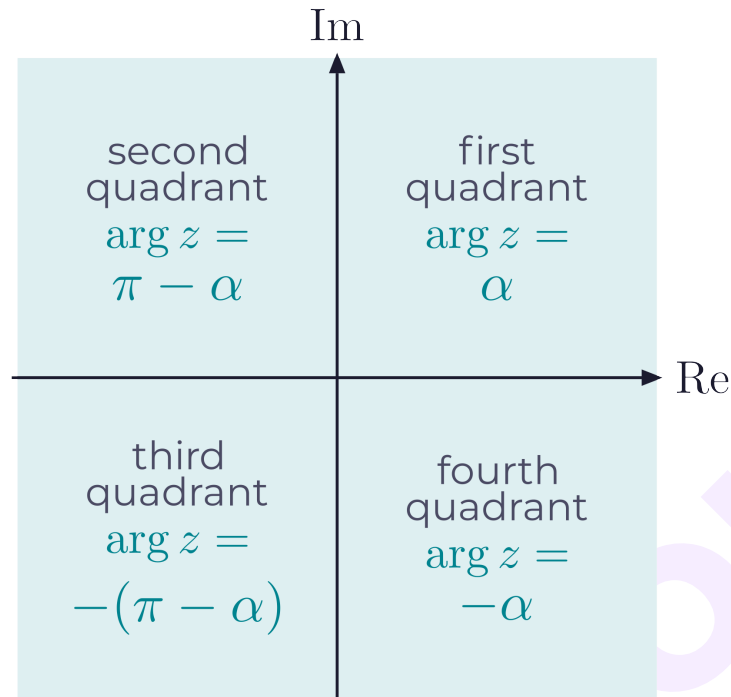
The same arrow can be reached by turning anticlockwise or clockwise, and by adding full turns of 2π , so one arrow has many angles. We agree to report one of them, the **principal argument**, which lies in $(-\pi, \pi]$: between 0 and π for an arrow above the real axis, and between $-\pi$ and 0 for an arrow below it. The shorter turn wins.

On the negative real axis the two turns are equally long, π anticlockwise and π clockwise. The agreement settles the tie in favour of the anticlockwise turn, so $\arg(-2) = \pi$, never $-\pi$.



Finding the argument in any quadrant

Work in two steps. First find the acute angle the arrow makes with the real axis, $\alpha = \tan^{-1} \left| \frac{y}{x} \right|$. Then read the argument from the quadrant the point is in. A point on an axis is in no quadrant, so read its argument straight from the picture: 0 on the positive real axis, π on the negative real axis, $\frac{\pi}{2}$ on the imaginary axis above the origin and $-\frac{\pi}{2}$ below it.



Key formulas

$$|z| = \sqrt{x^2 + y^2}$$

$$\alpha = \tan^{-1} \left| \frac{y}{x} \right|$$

first quadrant: $\arg z = \alpha$

second quadrant: $\arg z = \pi - \alpha$

third quadrant: $\arg z = -(\pi - \alpha)$

fourth quadrant: $\arg z = -\alpha$

polar form: $z = r(\cos \theta + i \sin \theta)$

with $r = |z|$, $\theta = \arg z$

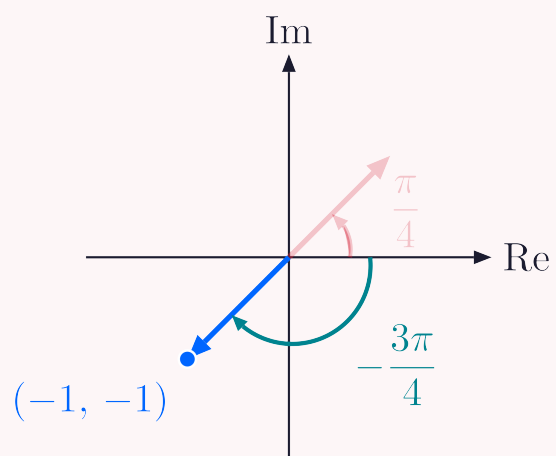
For the four arrows $1 + i$, $-1 + i$, $-1 - i$ and $1 - i$, the acute angle is $\alpha = \frac{\pi}{4}$ every time, and the arguments are $\frac{\pi}{4}$, $\frac{3\pi}{4}$, $-\frac{3\pi}{4}$ and $-\frac{\pi}{4}$.

CAREFUL

For $z = -1 - i$ a calculator gives

$$\tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} \left(\frac{-1}{-1} \right) = \tan^{-1}(1) = \frac{\pi}{4}.$$

But an arrow at $\frac{\pi}{4}$ points to $(1, 1)$ in the first quadrant, and $-1 - i$ is the point $(-1, -1)$ in the third. Dividing y by x threw away both signs. The true argument is $-(\pi - \frac{\pi}{4}) = -\frac{3\pi}{4}$. Always check the quadrant.



Solved example

Find $|z|$ and $\arg z$ for $z = -1 - \sqrt{3}i$.

Plot it. The point is $(-1, -\sqrt{3})$, in the third quadrant.

Modulus. $|z| = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{1+3} = 2$.

Acute angle. $\alpha = \tan^{-1} \left| \frac{-\sqrt{3}}{-1} \right| = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$.

Argument. Third quadrant, so $\arg z = -(\pi - \frac{\pi}{3}) = -\frac{2\pi}{3}$.

$$|z| = 2, \quad \arg z = -\frac{2\pi}{3}$$

Practice set

Easy to hard. Give every argument as the principal argument, in radians.

- Q1.** Find $|z|$ and $\arg z$ for $z = 1 + \sqrt{3}i$.
- Q2.** Find $|z|$ and $\arg z$ for $z = -3$.
- Q3.** Find $|z|$ and $\arg z$ for $z = -2i$.
- Q4.** Find $|z|$ and $\arg z$ for $z = \sqrt{3} - i$.
- Q5.** Find $|z|$ and $\arg z$ for $z = \frac{1+i}{1-i}$.
- Q6.** Write $z = -1 + \sqrt{3}i$ in polar form.
- Q7.** Find $|z|$ and $\arg z$ for $z = -4 - 4i$.
- Q8.** Let $z = -1 - i$. Find $\arg(z^2)$. Is it equal to $2 \arg z$?

Answers

- Q1.** $|z| = 2$, $\arg z = \frac{\pi}{3}$. $|z| = \sqrt{1+3} = 2$. First quadrant, $\alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$.
- Q2.** $|z| = 3$, $\arg z = \pi$. The point $(-3, 0)$ is on the negative real axis. The argument is π , not $-\pi$.
- Q3.** $|z| = 2$, $\arg z = -\frac{\pi}{2}$. The point $(0, -2)$ is straight down the imaginary axis: a quarter turn clockwise.
- Q4.** $|z| = 2$, $\arg z = -\frac{\pi}{6}$. $|z| = \sqrt{3+1} = 2$. Fourth quadrant, $\alpha = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$, so $\arg z = -\frac{\pi}{6}$.
- Q5.** $|z| = 1$, $\arg z = \frac{\pi}{2}$. Multiply top and bottom by $1+i$: $\frac{(1+i)^2}{(1-i)(1+i)} = \frac{2i}{2} = i$, the point $(0, 1)$.
- Q6.** $z = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$. $r = \sqrt{1+3} = 2$. Second quadrant, $\alpha = \frac{\pi}{3}$, so $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.
- Q7.** $|z| = 4\sqrt{2}$, $\arg z = -\frac{3\pi}{4}$. $|z| = \sqrt{16+16} = 4\sqrt{2}$. Third quadrant, $\alpha = \frac{\pi}{4}$, so $\arg z = -\frac{3\pi}{4}$, not $\frac{\pi}{4}$.

Q8. $\arg(z^2) = \frac{\pi}{2}$, **and no.** $z^2 = (-1-i)^2 = 1+2i+i^2 = 2i$, so $\arg(z^2) = \frac{\pi}{2}$. But $2 \arg z = 2 \left(-\frac{3\pi}{4}\right) = -\frac{3\pi}{2}$, which lies outside $(-\pi, \pi]$. The two differ by a full turn of 2π : they are the same direction, but only $\frac{\pi}{2}$ is the principal argument.



Watch the animated lesson on YouTube

Modulus and Argument of a Complex Number

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